

NARITA INTL, Japan

WMO: 476860

Lat: 35.765N

Lon: 140.386E

Elev: 43

StdP: 100.81

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.9	-2.8	-13.1	1.2	4.1	-11.2	1.4	3.9	11.5	7.2	10.2	6.7	1.7	330	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.4	33.0	26.0	31.8	25.6	30.2	25.1	26.7	30.7	26.1	29.9	25.6	29.2	4.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.9	21.3	29.5	25.1	20.3	28.3	24.2	19.3	27.4	84.4	31.1	81.5	29.9	78.9	29.1	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	8.7	7.6	DB	-6.5	35.0	1.1	1.1	-7.3	35.8	-8.0	36.5	-8.6	37.1	-9.5	37.9	(4)
(5)			WB	-7.4	28.2	1.0	0.7	-8.2	28.7	-8.8	29.2	-9.4	29.6	-10.1	30.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.8	3.8	4.7	8.0	13.0	17.5	20.8	25.0	26.2	22.9	17.5	11.6	6.4
	DBStd	8.19	2.36	2.83	3.35	3.37	2.69	2.58	2.78	2.32	3.06	2.91	3.11	2.92
	HDD10.0	576	193	152	82	11	0	0	0	0	0	0	19	119
	HDD18.3	1994	451	383	321	161	46	7	0	0	3	50	203	371
	CDD10.0	2346	1	3	20	103	234	325	465	504	387	233	66	6
	CDD18.3	722	0	0	0	2	22	82	207	246	139	25	1	0
	CDH23.3	5239	0	0	0	13	114	368	1664	2160	833	86	1	0
	CDH26.7	1677	0	0	0	1	10	67	583	787	218	11	0	0
Wind	WSAvg	3.6	3.4	3.7	4.0	4.1	3.8	3.4	3.5	3.4	3.7	3.6	3.1	3.1
Precipitation	PrecAvg	1452	66	65	106	114	130	149	132	122	200	203	105	61
	PrecMax	1961	148	127	192	206	213	283	279	286	399	599	249	193
	PrecMin	1041	1	9	43	42	63	44	16	14	35	26	7	2
	PrecStd	209	42	34	37	46	41	52	78	80	98	134	57	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.2	18.2	21.2	25.1	28.0	30.9	34.1	34.8	32.2	28.1	22.1	18.9
		MCWB	10.6	13.4	13.4	16.9	19.4	24.0	25.9	26.6	24.7	22.0	16.9	15.2
	2%	DB	12.9	14.8	18.8	22.8	26.0	28.2	32.9	33.1	30.9	25.8	20.1	15.9
		MCWB	8.4	11.0	12.8	15.9	19.0	22.3	25.8	26.2	24.8	21.0	15.9	11.5
	5%	DB	10.9	12.1	16.8	21.0	24.2	27.0	31.2	32.1	29.1	23.9	18.9	13.9
		MCWB	6.1	7.2	12.3	15.6	18.4	21.6	25.6	25.8	24.3	19.6	15.2	9.7
	10%	DB	9.1	10.2	14.8	19.1	23.0	25.2	30.0	30.9	27.8	22.1	17.1	12.1
		MCWB	4.6	5.9	10.6	14.1	17.7	21.2	25.1	25.7	23.6	18.5	13.8	8.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	14.5	16.0	20.0	22.4	24.8	27.6	28.0	26.3	23.6	19.0	16.4
		MCDB	14.3	17.4	18.0	21.7	25.2	29.1	32.1	32.7	29.5	26.1	20.8	18.4
	2%	WB	9.0	11.3	14.3	17.7	20.8	23.7	26.6	27.1	25.6	22.0	17.2	12.5
		MCDB	11.5	14.0	17.4	20.9	24.1	26.7	30.7	31.2	28.8	24.6	19.3	14.8
	5%	WB	6.9	8.5	12.9	16.3	19.7	22.9	25.9	26.4	25.0	20.7	15.7	10.4
		MCDB	9.5	11.1	16.1	19.5	22.7	25.6	30.0	30.2	28.1	22.9	17.7	12.6
	10%	WB	5.4	6.6	10.9	15.1	18.7	22.0	25.3	25.9	24.3	19.5	14.6	8.9
		MCDB	8.1	9.1	14.0	18.2	21.7	24.6	29.1	29.5	27.0	21.6	16.6	11.3
Mean Daily Temperature Range	5% DB	MDBR	10.3	9.8	9.9	9.6	8.8	7.1	7.3	7.4	6.9	7.6	9.3	10.2
		MCDBR	12.6	12.8	12.6	12.0	11.4	9.5	9.6	9.5	8.5	9.0	10.7	11.8
		MCWBR	8.7	9.0	8.6	7.1	6.1	4.7	4.2	3.9	4.0	5.5	7.3	8.3
	5% WB	MCDBR	10.7	11.7	11.4	9.8	9.4	7.9	8.8	8.5	7.7	7.7	9.0	10.3
		MCWBR	8.3	9.2	9.1	6.9	6.2	4.6	4.3	4.0	3.9	5.4	7.1	8.2
Clear-Sky Solar Irradiance	taub	0.379	0.430	0.498	0.547	0.562	0.589	0.583	0.564	0.502	0.476	0.426	0.383	
	taud	2.242	2.094	1.931	1.861	1.881	1.882	1.945	1.986	2.126	2.120	2.174	2.250	
	Ebn at Noon	795	791	767	754	749	727	727	732	758	736	732	762	
	Edh at Noon	107	139	178	201	200	199	186	175	145	134	114	100	
All-Sky Solar Radiation	RadAvg	2.69	3.15	3.99	4.68	5.05	4.53	4.97	5.01	3.85	2.96	2.43	2.35	
	RadStd	0.18	0.33	0.26	0.46	0.52	0.40	0.82	0.48	0.33	0.29	0.22	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.27	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page